Lab : Statistical Methods

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1 Discrete and Continuous Uniforms

Model Description

Computers are deterministic and cannot generate true continuous random variables. Instead, they generate pseudo-random continuous numbers by scaling large discrete integers. This module explores the mathematical convergence of a discrete uniform distribution to a continuous one as the state space expands, proving that continuous variables are effectively simulated via high-granularity discrete steps.

Implementation Task

1. **Discrete Generation:** Generate an array of discrete uniform integers $X \in \{0, 1, \ldots, N\}$ using `np.random.randint`.
2. **Linear Transformation:** Scale the discrete values into the standard uniform interval $[0, 1]$ using the transformation $U = \frac{X}{N}$.
3. **Observing Granularity:** Generate 10,000 samples for a small interval ($N = 10$) and plot the normalized histogram to visually identify the distinct discrete steps.
4. **The Continuous Limit:** Repeat the process for a massive integer range ($N = 2^{32} - 1$) to demonstrate how the discrete gaps visually vanish, merging into the perfectly flat continuous PDF $f(x) = 1$.
5. **Moment Verification:** Compute the empirical variance for both the small and large N cases. Compare these computationally derived values to the theoretical continuous variance $\frac{1}{12}$ to prove that the scaled discrete approximation converges.

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2 Generating Distributions via Inverse Transform

Model Description

Virtually all complex distributions are simulated by mathematically transforming a fundamental standard uniform variable $U \sim \text{Uniform}(0, 1)$. By leveraging the generalized inverse Cumulative Distribution Function (CDF), $F^{-1}(u) = \inf\{x \in \mathbb{R} : F(x) \geq u\}$, we can map a uniform probability space onto any target distribution.

Implementation Task

Part A: Continuous Transformation (Exponential)
Implement the algebraic inverse for the exponential distribution: $F(x) = 1 - e^{-\lambda x}$.

- Generate 5,000 samples for $\lambda = 2$.
- Overlay the theoretical PDF curve $f(x) = \lambda e^{-\lambda x}$ on top of your normalized empirical histogram to verify the transformation.

Part B: Discrete Transformation (Step-Function)
Implement the generalized inverse logic for a generic discrete distribution with probabilities $p_1 = 0.2, p_2 = 0.5, p_3 = 0.3$.

- Write a function utilizing `np.searchsorted` or `np.digitize` to map arrays of standard uniforms into their respective discrete buckets efficiently.
- Plot a bar chart of the resulting frequencies.

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3 Binomial Convergence & The Infinite Limit

Model Description

The Binomial distribution models the sum of n independent Bernoulli trials. By examining the sample proportion of successes ($\hat{p} = \frac{Y}{n}$), we can computationally validate the frequentist definition of probability. Mathematically, as the number of trials approaches infinity, the variance of the sample proportion collapses to zero:

$$\lim_{n \to \infty} Var(\hat{p}) = \lim_{n \to \infty} \frac{p(1-p)}{n} = 0$$

Implementation Task

1. **Simulation:** Simulate $Y \sim \text{Binomial}(n, p)$ by summing n independent Bernoulli trials.
2. **Rolling Variance:** Generate sample proportions $\hat{p}$ for logarithmically increasing values of n (from $n = 10$ to $n = 100,000$), holding the true probability p constant.
3. **Visualization:** Plot n on a logarithmic x-axis versus the empirical variance of $\hat{p}$ on the y-axis. Verify that your empirical scatter plot perfectly mirrors the theoretical decay curve $\frac{p(1-p)}{n}$.

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4 Discrete Convolution in Practice

Model Description

When X and Y are independent discrete random variables, the probability mass function of their sum $Z = X + Y$ is found via the discrete convolution of their individual PMFs:

$$p_Z(z) = \sum_x p_X(x) p_Y(z - x)$$

Implementation Task

1. Generate two independent discrete uniform variables X and Y (representing the rolling of two fair 6-sided dice) over 100,000 computational trials. Calculate the array of sums $Z = X + Y$.
2. Compute the theoretical PMF of Z without simulation by using `np.convolve` on the arrays of their individual theoretical probabilities.
3. Plot the empirical bar chart of your simulated Z values side-by-side with a stem plot of the exact mathematical convolution output to demonstrate their equivalence.